

DATE PRINTED 14/09/2017

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

VISUAL INSPECTION

| DATE       | STATUS  | OIL TEMP °C | W. TEMP °C | OIL LEVEL | PAINT | S. GEL | LEAKS |
|------------|---------|-------------|------------|-----------|-------|--------|-------|
| 04/07/2017 |         | 28          |            | NORMAL    |       | GOOD   |       |
| 31/07/2017 | ONLINE  |             |            |           |       |        |       |
| 30/08/2017 | OFFLINE | 35          |            | NORMAL    | FAIR  | GOOD   | NONE  |

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10031</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

LIQUID SCREEN TEST DATA

| DATE       | SERVICES | ASTM D974<br>ACID<br>(mgKOH/gm) | ASTM D971<br>IFT<br>(mN/m) | IEC 156<br>DIEL<br>(kV) | ASTM D1500<br>COLOUR | ASTM D4052<br>SP. GR. | ASTM D1524<br>VISUAL | ASTM D1698<br>® SEDIMENT |
|------------|----------|---------------------------------|----------------------------|-------------------------|----------------------|-----------------------|----------------------|--------------------------|
| 04/07/2017 | ROUTINE  | 0.048 AC                        | 28.4 QU                    | 44 AC                   | 1.0 AC               | 0.8537 AC             | CLEAR AC             |                          |
| 31/07/2017 | ROUTINE  | 0.042 AC                        |                            | 77 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |
| 30/08/2017 | ROUTINE  | 0.042 AC                        |                            | 61 AC                   | 1.5 AC               |                       | CLEAR AC             |                          |

DISSOLVED GAS ANALYSIS (ASTM D3612), EXPRESSED IN PPM

| DATE       | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON<br>MONOXIDE | CARBON<br>DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL<br>COMBUSTIBLE | TOTAL<br>GAS |
|------------|----------|--------|----------|---------|--------------------|-------------------|--------|----------|-----------|----------------------|--------------|
| 04/07/2017 | 32       | 15273  | 65111    | 5       | 95                 | 1002              | ND     | 2        | ND        | 134                  | 81520        |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - NO  |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

RECOMMENDATION:  
RETEST 12 MONTHS:  
TEST RESULT INDICATES NORMAL OPERATION.

WATER CONTENT (ASTM D1533), EXPRESSED IN PPM

| DATE       | TEMP°C | PPM | %M/dw | GRADE |
|------------|--------|-----|-------|-------|
| 04/07/2017 | 25     | 32  | 4.16  | E     |
| 31/07/2017 | 25     | 16  | 2.08  | D     |
| 30/08/2017 | 30     | 13  | 1.37  | C     |

FIELD SERVICE

| DATE | SERVICE |
|------|---------|
|------|---------|

DATE PRINTED 14/09/2017

ID# - 601669

CUST - PT IP UJP PLTU  
JERANJANG

SUB NAME -  
CITY - MATARAM LOMBOK  
UNIT# - **TRAFO 3 ESP UNIT 1**

LOCATION - OUTDOOR  
OTHER -

NAMEPLATE DATA

|                   |                |
|-------------------|----------------|
| MFG               | - FUJIAN       |
| DATE MFG          | - 2010         |
| MAIN TANK S/N     | - <b>10031</b> |
| KVA               | -              |
| PRI               | - 72000 V      |
| SEC               | - 380 V        |
| KILOGRAMS         | - 1000         |
| EQUIP TYPE        | - TRANSFORMER  |
| TRANS CLASS       | - ONAN         |
| IMPEDENCE         | -              |
| PHASE/CYCLE       | - 3/50 Hz      |
| LIQUID TYPE       | - OIL          |
| LITRES LIQUID     | - 630          |
| LTC MFG           | -              |
| LTC S/N           | -              |
| LTC FAB NO        | -              |
| LTC MODEL         | -              |
| LTC LITRES LIQUID | -              |
| LTC ENERGIZED     | -              |

ADDITIONAL EQUIPMENT

|                 |       |
|-----------------|-------|
| RADIATORS       | - YES |
| FANS            | - NO  |
| H2O COOLED      | - NO  |
| OIL PUMPS       | - NO  |
| TOP FPV         | -     |
| BOTTOM FPV      | -     |
| BUCHOLZ         | - NO  |
| CONSERVATOR     | - YES |
| LTC COMPARTMENT | - NO  |
| BUSHING LOC     | - TOP |
| FREE BREATHER   | - YES |
| HOSE LENGTH     | -     |
| POWER AVAILABLE | - YES |

FURAN ANALYSIS (ASTM D5837), EXPRESSED IN PPB

|            |      |      |      |      |      |       |
|------------|------|------|------|------|------|-------|
| DATE       | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
| 04/07/2017 | ND   | 32   | ND   | ND   | ND   | 32    |

RECOMMENDATION:  
FURAN ANALYSIS SHOWS DETECTION OF 2FOL WHICH INDICATES THAT THE SOLID INSULATION IS WET. IT IS ADVISABLE TO MAINTAIN THE OIL IN A GOOD CONDITION SO AS NOT TO FURTHER DETERIORATE THE PAPER INSULATION.

METALS-IN-OIL (ASTM D3635), EXPRESSED IN PPM

|            |            |       |        |
|------------|------------|-------|--------|
| DATE       | ®ALUMINIUM | ®IRON | COPPER |
| 04/07/2017 | 0.01       | 0.01  | 0.01   |

LIQUID POWER FACTOR (ASTM D924)

|            |          |          |
|------------|----------|----------|
| DATE       | 25°C     | 100°C    |
| 04/07/2017 | 0.000 AC | 1.700 AC |

INHIBITOR CONTENT, DBPC (ASTM D2668)

|            |            |
|------------|------------|
| DATE       | %BY WEIGHT |
| 04/07/2017 | 0.01 UN    |

®CORROSIVE SULFUR (ASTM 1275B)

|            |         |       |              |
|------------|---------|-------|--------------|
| DATE       | STATUS  | CLASS | APPEARANCE   |
| 04/07/2017 | NON-COR | 1A    | LIGHT ORANGE |

®PASSIVATOR COUNT, EXPRESSED IN PPM

|      |     |
|------|-----|
| DATE | PPM |
|------|-----|

FIELD SERVICE

|      |         |
|------|---------|
| DATE | SERVICE |
|------|---------|